

Education

University of California, San Diego | San Diego, CA

SEP 2024 - JUN 2026

M.S. in Computer Science (AI focus)

GPA: 3.90

Relevant Coursework: Deep Learning, Machine Learning, NLP, ML Systems, Artificial Intelligence, Recommender Systems, Computer Arch

University of California, Berkeley | Berkeley, CA

AUG 2020 - MAY 2024

B.A. in Computer Science & B.A. in Applied Math

GPA: 3.90

Professional Experience

Machine Learning Engineer Intern | Qualcomm

JUN 2025 -

- Architect multimodal RAG system through Langchain, using CLIP embeddings, LlamaCPP, FAISS vector store (8-bit quantization)
- Deploy M-RAG system on Qualcomm-native LeMans machine using Adreno GPU, benchmarking against CUDA host on Haystacks

Teaching Assistant | CSE 234: Data Systems for ML & CSE 153: ML for Music @ UCSD

DEC 2024 - JUN 2025

- Constructed 3 brand-new assignments for students covering mat-mul optimization, backprop graph construction, speculative decoding
- Hosted 20+ office hours, grade projects, proctor exams, hold discussion sections, maintain Ruby & Jekyll course website

Research Intern | Hao AI Lab @ UCSD

APR 2024 -

- Develop cutting-edge diffusion-based LLM (DLLM) specifically tuned on SQL and coding datasets
- Configured new CLLM models to be compatible with qLoRA fine-tuning on Jacobi trajectories, decreasing memory usage over 3x

Machine Learning Engineer Intern | Accretional

JUN 2024 - SEP 2024

- Designed and implement RAG vector embedding model to ground LLM with respect to cloud infrastructure
- Created command-line tool in Golang for indexing and vectorization of files, semantic & lexical search of keywords
- Fully constructed VSCode application in Svelte and TypeScript to help users easily implement and deploy APIs in less than 5 seconds

Software Engineer Intern | Hinge Health

MAY 2023 - AUG 2023

- Designed 5+ endpoints and event consumers in TypeScript and Ruby on Rails, restructuring and building new GraphQL queries
- Presented solutions for over 10 UI/UX features in React, linked repos with RabbitMQ event emitting, deployed in Kubernetes

Full Stack Intern | Cisco Meraki

MAY 2022 - AUG 2022

- Spearheaded a full stack project to create uplink ping stat graphs, helping 500,000 customers visualize wireless trends over time
- Collected ping stats data from over 4 million wireless internet nodes, using Scala scripts to build a grabber and Jenkins to test
- Conducted experiments with global LT & SQL queries across 600 shards to mitigate the size of rapidly-growing tables

Software Engineer Intern, Student Tutor | Breakout Mentors

JUN 2021 - AUG 2021

Projects & Research

MusicRFM, interpretable classification and tuned generation of music using recursive feature machines

JAN 2025 -

CoT Reasoning with Sparse Autoencoders

JAN 2025

Learner, a full stack Flask / React app that prompts & fine-tunes LLMs to help with learning languages

SEP 2024 - JAN 2025

Convolutional Neural Net, AlexNet, UNet implementation

MAR 2024 - APR 2024

- Engineered CNN, AlexNet, UNet and variants, including construction of layers, activations, autograd in NumPy, PyTorch

Insite, a React, NestJS & TypeScript app that connects high schoolers to professionals

JUN 2023 - SEP 2023

- Utilized AWS S3 buckets for file storage, assembled 40+ REST API endpoints for tasks like search, connections, and profile pages
- Built custom Docker containers and PostgreSQL databases, integrating them with CRA frontend & backend repos with networks

Regulus, a Golang secure file sharing system

MAR 2023 - MAY 2023

- Employed over 5 cryptographic tools and libraries like HashKDF, PBKDF, UUID, HMACs for authentication & access revocation

NumC, a C++ version of numpy matmul, optimized 40x using loop unrolling, SIMD, OpenMP, caches

OCT 2021 - DEC 2021

Skills

Python, SQL, JavaScript, TypeScript, React, PyTorch, TensorFlow, Java, Golang, Kubernetes, AWS, Ruby on Rails, Scala, Docker, C, C#, C++, RabbitMQ, GraphQL, HTML, CSS, Figma, Jenkins, PHP, Unity, Git / GitLab / GitHub, Unix, Jira, JupyterLab